

AI FITTRACK

PHASE 8 – PROJECT DEMONSTRATION

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1. DEMONSTRATION OVERVIEW

The Project Demonstration phase presents the completed AI FitTrack system through a structured screen-sharing demonstration. It shows the project name, purpose, uses and benefits, project execution, working process, and final output.

The demonstration should use the actual running project. Phase 6 contains detailed testing screenshots, while Phase 8 presents the live working flow.

2. DEMONSTRATION OBJECTIVES

Introduce AI FitTrack and its purpose.

Explain the major implemented capabilities.

Demonstrate server startup and MongoDB connection.

Demonstrate authentication and protected API access.

Demonstrate workout creation, retrieval, search, update, and deletion.

Demonstrate AI workout recommendation and fitness insights.

Show final API responses and working output.

Provide a clear conclusion.

3. DEMONSTRATION REQUIREMENTS

RequirementDemonstration Content

Screen sharingShow the project, terminal, project structure, and API testing interface.

Student voiceExplain the project and working process using the student's own voice.

Project nameAI FitTrack – Personalized Fitness Recommendations Powered by AI.

PurposeBackend REST API for authenticated workout management and AI-powered fitness guidance.

Uses / benefitsOrganized workout records, protected access, workout search, personalized recommendations, and fitness insights.

ExecutionShow server startup, database connection, and API execution.

Working processDemonstrate authentication, workout operations, search, and AI endpoints.

Final outputShow successful API responses and the final working system.

Video accessUpload the completed video to Google Drive and use an accessible link.

4. DEMONSTRATION SEQUENCE

Step Activity What to Explain:

1. Project Introduction Project name, purpose, problem addressed, and backend role.
2. Technology StackNode.js, Express.js, MongoDB, Mongoose, JWT, bcryptjs, and Google Gemini.
3. Project Structure src/config, controllers, middleware, models, routes, and services.
4. Environment and ExecutionStart the server and show MongoDB Atlas connection without exposing credentials.
5. Health Check Show the successful root endpoint response.
6. Registration Execute register and explain account creation.
7. Login Execute login and explain JWT authentication.
8. Profile Use JWT to access the protected profile endpoint.
9. Add Workout Create a workout and explain its fields.
10. Get All Workouts Retrieve the authenticated user's workout records.
11. Search Demonstrate search by name, category, and date.
12. Get by ID Retrieve a specific workout using its ID.
13. Update Modify an existing workout and show the response.
14. Delete the workout and demonstrate the post-delete state.
15. AI Recommendation Submit age, fitness goal, and experience and show the generated response.
16. AI Fitness InsightsSubmit total workouts, average duration, and total calories burned and show the generated insights.
17. Final Output Summarize successful modules and show final API output.
18. Conclusion Conclude the demonstration.

5. PROJECT INTRODUCTION CONTENT

AI FitTrack is a backend REST API designed for personalized fitness recommendations and workout management. The implemented system uses Node.js and Express.js for the API layer, MongoDB with Mongoose for persistence, JWT and bcryptjs for authentication and password protection, and Google Gemini for AI-powered fitness functions.

The demonstration should not claim a separate React frontend, administrator dashboard, additional fitness collections, or other features that are not part of the supplied implementation.

6. PURPOSE, USES AND BENEFITS

AreaDemonstration Explanation

PurposeProvide a structured backend service for authenticated workout management and AI-assisted fitness guidance.

Workout managementAllow authenticated users to create, retrieve, update, and delete workout records.

Workout discoveryAllow workout records to be searched by name, category, or date.

SecurityUse JWT authentication for protected API operations and bcryptjs for password hashing.

AI recommendationUse age, fitness goal, and experience as inputs for personalized workout guidance.

AI insights Use workout count, average duration, and calories burned as inputs for fitness performance insights.

Data persistence Store User and Workout data using MongoDB and Mongoose.

Maintainability Separate routes, controllers, middleware, models, configuration, and Gemini service responsibilities.

7. LIVE EXECUTION PROCEDURE

Before recording, confirm that the project starts successfully and MongoDB Atlas is available. Do not display the real .env file or any secret values.

Recommended execution order:

1. Open the AI-FitTrack project in Visual Studio Code.
2. Open the integrated terminal.
3. Start the application using the configured npm development/start command.
4. Show successful server startup.
5. Show successful MongoDB Atlas connection.
6. Open the root health endpoint.
7. Open Thunder Client and perform the API sequence in Section 4.
8. Show successful responses without exposing tokens or credentials.
9. Finish with AI recommendation and fitness-insights outputs.
10. Give the concluding explanation.

8. SCREEN AND VOICE PRESENTATION PLAN

Part Screen to Show Voice Explanation

Introduction Project title / workspace Name, purpose, and problem addressed.

Architecture Project structure / architecture diagram Explain client/API consumer, Express, routes, middleware, controllers, database, and Gemini service.

Execution Terminal Explain server startup and MongoDB connection.

Authentication Thunder Client Explain registration, login, JWT, and protected profile access.

Workout module Thunder Client Explain CRUD and search operations.

AI module Thunder Client Explain recommendation and insights inputs and outputs.

Conclusion Final response / workspace Summarize demonstrated functionality and scope.

9. SCREENSHOT AND VIDEO RELATIONSHIP

The screenshots stored under Phase 6 are evidence of individual test results. The Phase 8 video does not need to reproduce every screenshot as a slideshow. The presenter should show the live project and selected API responses while explaining the working process.

A live response demonstrated in the video may also correspond to a screenshot retained in Phase 6. This avoids duplicating the complete screenshot collection inside the demonstration document.

10. DEMONSTRATION CHECKLIST

CheckStatus Before Recording

Server starts successfully.

MongoDB Atlas connection succeeds.

Health endpoint responds successfully.

Register, login, and profile work.

Workout CRUD and search sequence works.

AI recommendation returns successful output.

AI fitness insights returns successful output.

No credentials or API keys are visible.

Screen sharing captures relevant windows.

Project name, purpose, uses and benefits are explained.

Final output is shown.

Google Drive link is added.

11. DEMO VIDEO LINK

The final demonstration video is to be recorded after the complete API and AI testing sequence has been verified. Upload the video to Google Drive with access configured for the evaluator.

Final Google Drive Demo Video Link:

https://drive.google.com/file/d/13T_CBNihCsZTN4DZon3Nsfi4b_BQWMJQ/view?usp=sharing

Do not insert an unverified public URL as the final submission link.

12. DEMONSTRATION CONCLUSION

The Project Demonstration phase provides the final presentation layer for AI FitTrack. It should show the running backend, explain purpose and benefits, execute the principal API operations, demonstrate AI functionality, and display final successful outputs.

After recording, insert the Google Drive video link in this document and add the link file to the Phase 8 GitHub folder. Then perform the final eight-phase repository and Skill Wallet check.